

Followup 26-04

New onset status epilepticus of undetermined cause

Patient	73-year-old female. COPD on 2 to 3 L home oxygen, generalized anxiety and major depression, diet-controlled type 2 diabetes, chronic low back pain on long-term opioids. No seizure history of any kind. Lives with her daughter and ambulates with a walker, managing her own diazepam and hydrocodone-acetaminophen while her daughter manages everything else.
Outcome	Survived. Three witnessed seizures before hospital arrival and one more in the emergency department, intubated for status epilepticus, four days of mechanical ventilation. CT head, MRI brain, and three days of continuous EEG never found a cause. The working diagnosis at transfer remained acute metabolic versus toxic encephalopathy, with benzodiazepine withdrawal the leading candidate.
Disposition	Extubated hospital day 4, transferred out of intensive care day 5, walking with a rolling walker and minimum assistance by day 5, discharge planning toward inpatient rehabilitation.

The call

Echo case. BLS took the call, managed her through a seizure on scene, and requested Echo. By the time she reached the hospital she had seized twice more.

BLS found her in a recliner, postictal, agitated, unable to answer questions or follow commands, moaning to a sternal rub, breathing at about 18 with adequate depth. Family reported she started shaking and stopped responding for about a minute, that she has no seizure history and nothing like this had happened before, that she had seemed confused and slow the day before, that she had taken a Norco the previous day, and that she had eaten very little for several days. No trauma, no falls.

Course, relative to BLS patient contact:

From contact	Event
+3 min	Vitals: pulse 130, respirations 18, SpO2 94%, glucose 167, GCS 8. Blood pressure unobtainable. Second seizure begins during the vital signs. Arms and legs extended, eyes rolled back. Moved to the floor, right lateral recumbent. Non-rebreather at 6 L. Lasts about a minute.
+8 min	Cardiac monitor applied by BLS during a 12 lead attempt.
+10 min	Non-rebreather increased to 10 L.
+12 min	Echo patient contact. Airway patent, lungs clear bilaterally, skin normal. Unresponsive, GCS 3. Both pupils dilated.
+16 min	IV, 18 gauge, right antecubital.
+17 min	12 lead: sinus tachycardia, no ST segment abnormality.
+18 min	145/109, MAP 121, pulse 130, respirations 18, SpO2 95%, glucose 176, GCS 3.

From contact	Event
+20 min	Oxygen reduced to nasal cannula at 2 L.
+25 min	Rolled onto a transfer sheet. She wakes, confused and uncooperative, repeatedly pulling off the oxygen mask and the monitor cables. Carried out to the cot.
+29 min	Third seizure just before departure, clonic, lasting about 90 seconds.
+31 min	Midazolam 5 mg IV. Seizure stops. Unresponsive again.
+33 min	140/68, MAP 92, pulse 138, SpO2 84%, ETCO2 50. GCS 3, charted with the chemically sedated qualifier.
+33 min	Nasal cannula replaced with non-rebreather at 10 L.
+40 min	110/50, MAP 70, pulse 115, SpO2 97%, ETCO2 23. GCS 3.
+53 min	At the receiving facility.

En route the Echo crew documented no movement of the arms or legs, both feet extended and rigid, pupils still dilated, no response to painful stimulus, and a patient still protecting her own airway and breathing at adequate rate and depth.

Hold onto the two capnography readings: **50, then 23, seventeen minutes apart**. Both come back in the labs, and they are not measuring the same thing.

Primary impression

Seizure, then postictal, then seizure again.

That is right, and there is not a fourth thing it could have been. She was witnessed convulsing by family and by both crews. What neither crew could have known, and what the hospital spent five days failing to establish, is why.

Three seizures without a return to baseline in between is status epilepticus, and the emergency department entered exactly that diagnosis. The definition most people carry from class is five minutes of continuous convulsing. The other half of it is the one that applied here: repeated seizures with incomplete recovery between them. She never got back to baseline, and she was in status from the recliner onward.

Hospital course

Emergency department, day 0. GCS 5 on arrival, labored breathing, not returning to baseline. Rapid sequence intubation with etomidate and succinylcholine, first pass, grade I view, 8.0 tube at 21 cm, fully confirmed. Sedated with ketamine, fentanyl, and propofol. Levetiracetam 3 g IV, with one further brief episode of seizure-like activity and **decerebrate posturing** before it was in. No hypoglycemia. CT head showed no hemorrhage, hydrocephalus, or herniation. Lactated Ringer's 1 L. Acetaminophen and salicylate negative, urine drug screen positive for benzodiazepines and opiates, urinalysis unremarkable.

Day 1. Potassium 2.7 and hemolyzed, repleted. The first arterial gas after intubation showed a **pH of 7.70** with a PCO2 of 22; the ventilator rate was cut from 16 to 12 and the tidal volume from 500 to 400, and the repeat gas came back at 7.47. Continuous EEG started. Lactate had fallen from 24 to 1.6 and creatinine from 1.54 to 1.04.

Days 2 and 3. Hypotension overnight unresponsive to two boluses, norepinephrine started to hold a MAP above 60 and off again the following day. She opened her eyes to voice but followed no commands. EEG still showed no seizures, and was discontinued so MRI could be obtained.

Day 4. MRI showed a very small area of restricted diffusion in the right parietal cortex, read as probable cortical superficial siderosis from remote trauma or a small vascular abnormality, and unlikely to be a seizure focus. Off sedation she was obtunded, opening her eyes to voice inconsistently, following no commands, withdrawing briskly to central pain. Extubated in the afternoon to 2 L. Neurology signed off with levetiracetam twice daily, repeat MRI in three months, and outpatient follow-up.

Day 5. Alert, following commands, asking repeatedly to go home. Transferred out of intensive care and walking with a front-wheeled walker at minimum assistance.

Final diagnoses: status epilepticus, resolved; new onset seizure disorder of undetermined etiology; acute metabolic versus toxic encephalopathy; acute on chronic hypoxic respiratory failure; lactic acidosis, resolved; acute kidney injury, resolved; hypokalemia, resolved; reactive leukocytosis, resolved; undifferentiated shock, resolved.

Education points

What a generalized seizure does to the rest of the body

A seizure is a brain event, and almost everything abnormal in this patient's labs is a muscle event.

In a generalized tonic-clonic seizure every skeletal muscle contracts at maximum effort against other muscles doing the same, with no rest between. That is more work than sprinting, and it outstrips what the circulation can deliver, so muscle switches to anaerobic metabolism. The byproduct is **lactate**. At the same time she was not ventilating properly, so delivery was falling while demand sat at its ceiling.

Her lactate was **24.0**. For scale, the septic shock threshold most people carry is 4, and 24 is a number you expect in a patient who is dying. She was not. **A high lactate tells you a large amount of tissue recently ran out of oxygen. It does not tell you why, and it does not tell you the tissue is still in trouble.** Seizure lactate peaks within minutes and clears in an hour or two once the muscles stop, because the liver takes it straight back to glucose. Hers went from 24 to 1.6 overnight. That pattern is reliable enough that emergency departments use lactate to separate a true generalized seizure from a faint or a psychogenic event.

Everything else follows from the same event. **Potassium 3.1** and later 2.7, because the catecholamine surge drives potassium into cells through beta-2 receptors, on top of days of poor intake. **White count 17.1**, normal within two days on no antibiotics, which is demargination rather than infection. **Glucose 254** against an A1c of 5.7, which is stress hyperglycemia and not diabetes. **Creatinine 1.54** falling to 1.04. **Troponin 19** from demand. None of it needed treatment. All of it was the seizure, written into the chemistry.

Benzodiazepine withdrawal

This is the diagnosis the hospital kept coming back to and never proved, and it is worth understanding because it is invisible from the outside and it is common in exactly this patient.

GABA is the brain's main inhibitory neurotransmitter. It opens a channel that lets chloride into the neuron, making the neuron harder to fire. **Benzodiazepines do not create inhibition.** They bind the GABA-A receptor and make it more responsive to the GABA already there.

Take that drug every day for years and the brain compensates by reducing the number and sensitivity of those receptors. The patient reaches a new equilibrium in which normal function requires the drug, while excitatory signalling, mostly glutamate, is unchanged and now sits opposite a weakened brake. Take the drug away and nothing holds the brake down. That produces anxiety, tremor, autonomic instability, insomnia, and at the far end, **seizures**. Diazepam has a long half-life and active metabolites, so withdrawal from it is delayed and can begin days after the last dose rather than hours.

Now put this patient in that frame. She self-manages her own diazepam, she had eaten very little for several days, and she had been confused and slow the day before. Her daughter administers everything else but not that one. Her urine screen was positive for benzodiazepines, which sounds like it rules withdrawal out and does not, because metabolites stay detectable for days to weeks after the last dose, and detectable is not the same as an adequate daily amount in someone who is dependent.

One more piece worth sitting with. **Midazolam stopped her seizure**. A benzodiazepine is first line for any seizure, so on its own that proves nothing. But in a withdrawal seizure it is simultaneously the correct emergency treatment and the specific antidote, and the hospital then held her home diazepam for five days because it was sedating her.

None of this was ever established. I am not telling you she had benzodiazepine withdrawal. I am telling you it stayed at the top of a differential through five days of imaging and EEG.

Further reading on withdrawal. The receptor pharmacology behind this takes some more studying to understand well. StatPearls has a free, readable chapter covering benzodiazepine, alcohol, and opioid withdrawal side by side, including why the GABA downregulation produces seizures: <https://www.ncbi.nlm.nih.gov/books/NBK459239/>

Posturing, and what it does not always mean

Three separate descriptions of this patient exist in the record: no movement in the arms or legs with both feet extended and rigid and no response to painful stimulus, convulsing with extended arms and legs, and seizure-like activity with decerebrate posturing. Those are the same phenomenon and only one of them names it. **Decerebrate posturing** is extension: arms extended and rotated inward, legs extended, feet pointed. **Decorticate posturing** is flexion, arms drawn in toward the core. The difference is anatomic. Above the red nucleus in the midbrain the pathway driving arm flexion is intact and you get decorticate; below it only the extensor drive from the vestibular nuclei remains, and you get decerebrate. Decerebrate therefore indicates a lower and worse level of brainstem involvement, and both are taught as signs of a mass, a bleed, or herniation.

Her CT was clean and her MRI was essentially clean. **She was posturing without a structural lesion**, because severe metabolic derangement and the immediate postictal state can both suppress higher cortical function with nothing occupying space in the skull.

Posturing still matters and belongs in your report every time. The harder half is that extensor posturing and tonic seizure activity look very similar from a few feet away, and both crews here described the same movements while the event was being called a seizure. Describing exactly what you saw, which is what both crews did, beats picking a label.

Prehospital considerations

Get the medication list, and ask who manages which pills. Nobody in that apartment could produce a list. The most useful sentence in this chart came from the daughter at the hospital: she manages her mother's medications except the diazepam and the Norco. That sentence put withdrawal at the top of the differential, and it is a question you can ask on scene.

GCS is a snapshot of a moving target. She was documented at 8 by BLS, 3 by Echo, and 5 in the emergency department inside of an hour, and nobody was wrong. A postictal level of consciousness changes minute to minute, and she woke combative in the middle of it and then seized again. Chart the number with the time beside it and expect the receiving team to get a different one.

A first seizure at 73 is not epilepsy until proven otherwise. It is a symptom of something else, and that is what everybody downstream will hunt for.

The interventions

PREHOSPITAL INTERVENTIONS

Oxygen, titrated four times. Non-rebreather at 6, then 10, then nasal cannula at 2, then non-rebreather at 10 again. That looks busy on paper and every change was right. Her home baseline is 2 to 3 L, so coming down once she was saturating in the mid-nineties made sense, and going back up when she hit 84% after the midazolam made sense. Worth stating plainly because the old teaching lingers: **in an acutely hypoxic patient you give oxygen, including a COPD patient.** The worry about suppressing respiratory drive belongs to a chronic hypercapnic patient on prolonged high-flow oxygen, not a hypoxic 73-year-old in status. Fix the hypoxia, then titrate. BLS also put her right lateral recumbent before Echo arrived, which deserves its own mention: a postictal patient has no gag protection and frequently vomits, and lateral recumbent lets the airway drain rather than pool while keeping the tongue off the posterior pharynx.

Midazolam 5 mg IV. Benzodiazepines are first line for seizure termination because they do the opposite of what withdrawal does: they enhance GABA-A signalling and restore inhibition. It worked in about the time it takes to say so.

Then look at what it cost, on the crew's own monitor two minutes later. **SpO2 84%, ETCO2 50.** That is hypoventilation, and it is not a complication, it is the drug. Benzodiazepines depress the brainstem respiratory centers along with everything else, and that depression deepens with dose, age, and existing lung disease. She had all three. **You are trading a seizure for a period of hypoventilation, and you have to be ready to own the airway when you make the trade.** This crew was watching the right numbers, caught it, and treated it with oxygen.

IN-HOSPITAL INTERVENTIONS

Rapid sequence intubation with etomidate and succinylcholine. Etomidate is chosen in unstable patients because it sedates with very little effect on blood pressure. **Succinylcholine** is a depolarizing paralytic: it binds the acetylcholine receptor at the neuromuscular junction and holds it open, producing a brief burst of fasciculation and then complete paralysis lasting only a few minutes. That short duration is the point, because it lets you take one look and hand the patient back their own breathing if it fails. The caution behind it is that holding those channels open releases potassium out of muscle cells, and in crush injury or established rhabdomyolysis that rise can be dangerous. She had just run every skeletal muscle in her body to exhaustion, which is why the department checked a creatine kinase. It was normal.

Levetiracetam 3 g load, then 750 mg twice daily. After a benzodiazepine stops a seizure, a longer-acting agent stops the next one. Levetiracetam binds a protein called SV2A on synaptic vesicles and reduces excitatory neurotransmitter release. It is the usual second line choice because it is available intravenously, has almost no drug interactions, needs no blood levels, and does not drop blood pressure.

Continuous video EEG, for three days. The reason to run it for days rather than take one recording is **nonconvulsive status epilepticus**: a brain still seizing electrically in a body that has stopped moving. In a patient who does not wake up after status, that has to be excluded before anything else is accepted. Hers showed no seizures across three days. What it did show was generalized slowing and rhythmic delta activity, the signature of **diffuse encephalopathy**, a whole brain working badly rather than one part misfiring. That pointed away from epilepsy and toward metabolic or toxic causes, where the diagnosis stayed.

Norepinephrine to a MAP above 60, started on day 2 after fluid failed. The number is worth noting because a physician here explicitly accepted 60 for this patient rather than the 65 that gets quoted almost everywhere. That figure is a widely used default, not a law of nature.

The labs, and what they mean

Reference ranges vary between institutions and analyzers. Several values below were reported in the record without a printed range, and standard adult ranges have been supplied and marked.

The acid-base arc, from the recliner to the ventilator

Lab	ED, day 0	Day 1, first gas	Day 4	Reference range
ETCO2, field (mmHg)	50, then 23	not applicable	not applicable	35 to 45 (supplied)
pH	7.34	7.70	7.44	7.35 to 7.45 (supplied)
PCO2 (mmHg)	41	22	47	35 to 45 (supplied)
Bicarbonate, serum (mmol/L)	9	27	31.8	22 to 29 (supplied)
Anion gap (mmol/L)	35	not repeated	13	5 to 14 (supplied)
Lactate (mmol/L)	24.0	1.6	not repeated	0.5 to 2.0 (supplied)
Base excess (mmol/L)	-3.7	not reported	6.7	-2 to +2 (supplied)

Follow this one all the way through, because every stage explains the next.

Stage one, the acid. Her muscles produced a lactate of 24. Buffering that acid consumed her bicarbonate, which is why serum bicarbonate came back at 9. The anion gap of 35 is the arithmetic confirmation: sodium minus chloride minus bicarbonate, elevated because lactate is an unmeasured anion sitting in that gap. Twenty-four points of lactate on top of a normal gap of about ten gives thirty-four. The numbers agree exactly.

Stage two, her own compensation, which the crew measured. The fastest response to a metabolic acid load is to blow off CO2, because CO2 in solution is an acid. Expected PCO2 is one and a half times the bicarbonate plus eight, so a bicarbonate of 9 predicts a PCO2 of about 21 to 22. The field capnography read 23, with a spontaneous respiratory rate the emergency department documented at 33.

That number was never an airway problem. It was a 73-year-old woman doing exactly the right thing, breathing more than thirty times a minute to hold her pH at 7.34 against a bicarbonate of 9. Without it, that bicarbonate with a normal PCO2 of 40 gives a pH near 7.0. **Her hyperventilation was the only thing between her and a pH incompatible with a working heart**, and the crew has the number that documents it.

Stage three, the other reading. Seventeen minutes earlier that same monitor read 50, with a saturation of 84%, two minutes after 5 mg of midazolam. Same patient, same device, opposite direction. Fifty was the drug

suppressing her respiratory drive; twenty-three was her metabolism demanding it back. **A capnography number means nothing without knowing what just happened to the patient.**

Stage four, the ventilator gets it wrong. She was set at a rate of 16 and a tidal volume of 500, a reasonable adult starting point and well below what she had been doing for herself. But the ventilator does not know the lactate is clearing. Her liver converted it back to glucose over the following hours, the acid went away, and the machine kept ventilating as though it were still there.

The result was a **pH of 7.70**. Severe respiratory alkalosis is not benign: it constricts cerebral arterioles and cuts cerebral blood flow, it shifts the oxyhemoglobin curve so hemoglobin holds oxygen more tightly instead of releasing it, and it drives potassium into cells, which matters in a patient already at 2.7. The team caught it, cut the rate to 12 and the volume to 400, and the pH came back to 7.47.

Stage five, the kidney's slow answer. By day 4 her bicarbonate was 31.8 with a base excess of +6.7, both high, in a patient with no acid left. That is renal compensation overshooting, because the kidney spent day 0 retaining every bicarbonate it could and takes days to change its mind. Her pH was 7.44, entirely normal, held there by a PCO2 of 47 that was also above range. **Two abnormal numbers producing one normal one.**

Potassium, kidney, and the rest of the panel

Lab	Day 0	Day 1	Day 4	Reference range
Potassium (mmol/L)	3.1	2.7	4.5	3.5 to 5.1 (supplied)
Sodium (mmol/L)	132	not reported	138	136 to 145 (supplied)
Chloride (mmol/L)	88	not reported	98	98 to 107 (supplied)
Creatinine (mg/dL)	1.54	1.04	0.93	0.60 to 1.10 (supplied)
eGFR (mL/min/1.73 m2)	35	not reported	65	60 or greater (supplied)
Glucose (mg/dL)	254	not reported	103	70 to 99 (supplied)
Hemoglobin A1c (%)	5.7 (prior year)	not applicable	not applicable	Less than 5.7 (supplied)
White blood cell (x10 ⁹ /L)	17.1	12.0	8.5	4.0 to 10.8 (supplied)
hs troponin T (ng/L)	19	not reported	not reported	Less than 15 (supplied)
Lipase (U/L)	109	not reported	not reported	13 to 60 (supplied)
Creatine kinase	Normal, value not reported	not repeated	not repeated	26 to 192 (supplied)

The potassium is the counterintuitive one. Acidosis usually raises serum potassium, because hydrogen ions moving into cells push potassium out. She had one of the most severe acidoses you will ever see in a chart, and her potassium was **3.1**, then **2.7**. That shift is much weaker in a lactic acidosis than in a mineral acid load, because lactate crosses the cell membrane along with the hydrogen ion instead of leaving potassium behind to balance the charge. And more powerfully, the catecholamine surge of a generalized seizure acts on beta-2 receptors and drives potassium **into** cells. Add days of poor intake and the patient is genuinely depleted. **A low potassium in the middle of a severe acidosis means the total body deficit is worse than the number looks**, because the acidosis is propping the measured value up.

Creatinine 1.54 with an eGFR of 35, back to 1.04 the next day. Acute kidney injury from the acidosis and hypoperfusion of the event, resolving as fast as it appeared. A creatinine of 1.54 is not dramatic on its own, and

the only reason anyone can call it an injury is that the later values exist to compare against. **Chloride 88** is the interesting half of the electrolytes, because chloride and bicarbonate move in opposite directions to keep charge balanced, so a chloride that low is the mirror image of a bicarbonate that got consumed and a gap that filled with lactate.

Two values worth reading on your own. Her **lipase was 109**, roughly double the top of the range. Lipase rises in pancreatitis but also in renal impairment, which she had, since the kidney clears it. Her **troponin was 19** against a female limit of 15, a type 2 elevation from an hour of tachycardia and maximal muscle work. Neither needed action, and neither changed her course. Both are worth being able to explain, because working out why a mildly abnormal value is there is the skill this section teaches.

Learning points

1. **Capnography measures metabolism, not just ventilation.** Her ETCO₂ read 50 and then 23 within seventeen minutes on the same monitor. The 50 was a benzodiazepine suppressing her respiratory drive. The 23 was her body blowing off CO₂ to defend a pH against a lactate of 24, matching the predicted compensation for a bicarbonate of 9 almost exactly. A low ETCO₂ is not automatically hyperventilation or poor perfusion, and a rising one is not automatically a failing airway.
2. **A high lactate immediately after a convulsion is expected, and it is not shock.** Every skeletal muscle just worked at maximum against no oxygen supply. Hers was 24 and cleared to 1.6 overnight. The same number without a convulsion means something entirely different, and a lactate that stays up after a seizure means a second problem.
3. **New onset seizures in an older adult on chronic benzodiazepines are withdrawal until proven otherwise.** Chronic use downregulates GABA-A receptors, and removing the drug leaves excitation unopposed. Diazepam's long half-life delays onset by days, and a positive urine screen does not rule it out, because metabolites persist long after the last adequate dose. Ask who fills the pillbox, because that answer is frequently the diagnosis.
4. **Posturing tells you how badly the brain is working, not why.** Decerebrate is extension, below the red nucleus; decorticate is flexion, above it. Both are taught as signs of herniation, and this patient postured with a clean CT and MRI, because severe metabolic derangement and the postictal state can produce it with no mass at all. Describe what you saw and do not let the label pick the cause.
5. **When you give a benzodiazepine for a seizure, you have bought an airway problem.** It is still the right drug and it was the right dose. Respiratory depression is the expected pharmacologic consequence, not a complication, and it arrives inside a couple of minutes, harder in an older patient with lung disease. Have oxygen and an airway plan ready before the drug goes in.

Prepared from records obtained through the follow-up request process and de-identified for education. To request follow-up on a call you ran, fill out the request form at <https://bcmca.github.io/followup/>